

FIDELITY OF QUANTUM STATES

Talk 3 - Jul. 9/26

Review:

- classical discrimination of a classical r.v. x drawn from pdf's $p(x)$ or $q(x)$.

Rule: Given observation x choose the greater of

$$\begin{cases} p(x) \longrightarrow x \sim p \\ q(x) \longrightarrow x \sim q \end{cases} \quad \leftarrow \text{gives maximum prob. of success.}$$

- quantum discrimination between two mixed states: ρ, σ .
- we designed the optimal POVM: $\{E, I-E\}$
- the prob. of successful guessing in both cases is:

$$p_{\text{classical}}^{\text{succ}} = \frac{1}{2} \left(1 + \frac{1}{2} \|\rho - \sigma\|_1 \right)$$

$$p_{\text{quantum}}^{\text{succ}} = \frac{1}{2} \left(1 + \frac{1}{2} \|\rho - \sigma\|_1 \right)$$

$$\text{where } \|A\|_1 = \sum_i |\lambda_i(A)|$$

$$= \text{tr} \sqrt{A^\dagger A}$$

- defined the trace distance btwn states ρ & σ :

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1$$

- is an operational defⁿ in terms of a classification task
→ also a nice mathematical defⁿ & interpretation as a proper metric.

→ other operational defⁿs are possible and useful in their own particular applications

⇒ Today we look at fidelity, $F(\rho, \sigma)$, btwn two states.

- ②
- Fidelity is a generalization of overlap between two pure states: $\langle \psi | \phi \rangle$
 → want something similar for mixed states.

Claim:

For pure states there is a direct relation btwn overlap and distinguishability:

$$\langle \psi | \phi \rangle \sim \frac{1}{2} \| |\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| \|_1 \quad (\text{not equal!})$$

Why?

- take 2-dim subspace spanned by $|\psi\rangle$ & $|\phi\rangle$.
- take basis vector $|e_1\rangle \equiv |\psi\rangle$ and any o.n. $|e_2\rangle$.

$$\Rightarrow |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\phi\rangle = \cos\theta |e_1\rangle + \sin\theta |e_2\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

where $\langle \psi | \phi \rangle = \cos\theta$.

$$\text{- so } |\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} c & s \\ s & s \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} c^2 & cs \\ cs & s^2 \end{pmatrix}$$

$$= \begin{pmatrix} s^2 & -cs \\ -cs & -s^2 \end{pmatrix} \rightarrow \text{need } \sum_{i=1}^2 |\lambda_i| \text{ of eigenvalues}$$

- recall Cayley's th^m for 2×2 matrices: $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$.
- $\Rightarrow$ here: $\text{tr} = 0$

$$\det = -s^4 - c^2 s^2 = -s^2(s^2 + c^2) = -\sin^2\theta.$$

$$\therefore \lambda - \sin^2\theta = 0$$

$$\Rightarrow \lambda = \pm \sin\theta.$$

$$\therefore |\lambda_1| + |\lambda_2| = 2 \sin \theta$$

$$\text{So, } \frac{1}{2} \| |\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| \|_1 = \sin \theta$$

$$\text{But, } |\langle\psi|\phi\rangle|^2 = \cos^2 \theta.$$

$$= T(|\psi\rangle, |\phi\rangle)$$

← simple 1:1 relation.
for pure states.

- this squared-overlap is what we call fidelity for pure states.

General Defⁿ: Fidelity.

- for two mixed states in general, ρ, σ :

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|^2$$

Claim: This recovers the pure state definition above: $\cos^2 \theta$.

Proof: Take $\rho = |\psi\rangle\langle\psi|$, $\sigma = |\phi\rangle\langle\phi|$.

$$\text{Want to show } \|\sqrt{\rho}\sqrt{\sigma}\|^2 = |\langle\psi|\phi\rangle|^2$$

- note for rank arbitrary projections, like $\Pi = |\psi\rangle\langle\psi|$,
 $\Pi^2 = \Pi$

$$\therefore \Pi = \sqrt{\Pi}$$

$$\text{• so, } F(\psi, \phi) = \|\sqrt{|\psi\rangle\langle\psi|} \sqrt{|\phi\rangle\langle\phi|}\|^2$$

$$= \|\underbrace{|\psi\rangle\langle\psi|}_{\langle\psi|\phi\rangle \in \mathbb{C}} |\phi\rangle\langle\phi|\|^2$$

$$\langle\psi|\phi\rangle \in \mathbb{C}$$

$$= \|\langle\psi|\phi\rangle \cdot |\psi\rangle\langle\psi|\|^2 \rightarrow \text{but } \|cA\| = |c| \|A\|,$$

$$= |\langle\psi|\phi\rangle|^2 \|\underbrace{|\psi\rangle\langle\psi|}_{=1}\|^2 \rightarrow = 1? \text{ (YES!).}$$

↳ answer we are expecting

- now need the one-norm of rank-one operator $|\psi\rangle\langle\phi|$.

• take $X = |\psi\rangle\langle\phi|$

$$\rightarrow \|X\|_1 = \text{tr} \sqrt{X^\dagger X} \text{ from defn.}$$

$$\begin{aligned} \text{• but } X^\dagger X &= |\phi\rangle\langle\phi| \cdot \underbrace{|\psi\rangle\langle\psi|}_{=1} \\ &= |\phi\rangle\langle\phi| \end{aligned}$$

$\therefore \sqrt{X^\dagger X} = |\phi\rangle\langle\phi|$ since it's a rank-1 projection.

$$\begin{aligned} \therefore \text{tr} \sqrt{X^\dagger X} &= \text{tr} |\phi\rangle\langle\phi| \\ &= \text{tr} \langle\phi|\phi\rangle \\ &= \text{tr} 1 \\ &= 1. \end{aligned}$$

$$\therefore \| |\psi\rangle\langle\phi| \|_1 = 1.$$

$$\therefore F(|\psi\rangle, |\phi\rangle) = |\langle\phi|\psi\rangle|^2 \text{ as expected.}$$

• what about a pure & a mixed state: $|\psi\rangle$ & ρ ?

Claim: $F(|\psi\rangle, \rho) = \langle\psi|\rho|\psi\rangle$

Proof:

$$\begin{aligned} \text{• We have } F(\psi, \rho) &= \|\sqrt{|\psi\rangle\langle\psi|} \sqrt{\rho}\|_1^2 \\ &= \| |\psi\rangle\langle\psi| \sqrt{\rho} \|_1^2 \quad (\because \text{projection}) \\ &= \|\pi \sqrt{\rho}\|_1^2 \end{aligned}$$

• but $\|A\|_1 = \text{tr} \sqrt{AA^\dagger}$

$$\begin{aligned} \therefore \|\pi \sqrt{\rho}\|_1 &= \text{tr} \sqrt{\pi \sqrt{\rho} \sqrt{\rho} \pi} \\ &= \text{tr} \sqrt{\pi \rho \pi} \end{aligned}$$

• but $\pi \rho \pi = |\psi\rangle\langle\psi| \rho |\psi\rangle\langle\psi|$

$= \langle\psi|\rho|\psi\rangle \cdot |\psi\rangle\langle\psi|$

$\therefore \text{tr} \sqrt{\pi \rho \pi} = \sqrt{\langle\psi|\rho|\psi\rangle} \text{tr} \sqrt{|\psi\rangle\langle\psi|}$
 $= \sqrt{\langle\psi|\rho|\psi\rangle} \underbrace{\text{tr} |\psi\rangle\langle\psi|}_{=1}$
 $= \sqrt{\langle\psi|\rho|\psi\rangle}$

$\Rightarrow \|\psi\rangle\langle\psi| \rho\|_1 = \sqrt{\langle\psi|\rho|\psi\rangle}$

$\therefore F(\psi, \rho) = \langle\psi|\rho|\psi\rangle$

Both cases have a similar operational interpretation of prob. of measurement outcome with POVM:

POVM: $\{|\psi\rangle\langle\psi|, I - |\psi\rangle\langle\psi|\}$

$F(\psi, \phi) = |\langle\psi|\phi\rangle|^2 = \text{prob. outcome } \psi \text{ when given state } |\phi\rangle$

$F(\psi, \rho) = \langle\psi|\rho|\psi\rangle = \text{" " } \psi \text{ " " " } \rho$

What about mixed states?

$F(\rho, \sigma) = \|\sqrt{\rho} \sqrt{\sigma}\|_1^2$

• for commuting operators its easy to show

$F(\rho, \sigma) = \left(\sum_i \sqrt{p_i q_i} \right)^2$ (when $p_i \sim \text{eigenvals}(\rho)$
 $q_i \sim \text{eigenvals}(\sigma)$)
 $= \text{Bhattacharyya distance of } \{p_i\}, \{q_i\}$
 as classical prob. dist's.

Non-Commuting Mixed States

How to measure an "overlap" of mixed states?

→ recall purification of mixed states

- take a mixed state ρ_A on a system A
- add a reference system R and produce a pure state $|\psi\rangle_{RA}$ on the joint system RA such that discarding (tracing out) R on the joint state produces ρ_A .

$$\Rightarrow \rho_A \xrightarrow{\text{purification}} |\psi\rangle_{RA}$$

$$\text{s.t. } \text{tr}_R |\psi\rangle_{RA} \langle \psi|_{RA} = \rho_A$$

- do this purification on both $\rho, \sigma \Rightarrow \rho \rightarrow |\psi_\rho\rangle$
 $\sigma \rightarrow |\psi_\sigma\rangle$
- try now $F(\rho, \sigma) \stackrel{?}{=} |\langle \psi_\rho | \psi_\sigma \rangle|^2$

→ not quite!

Turns out this is not unique.

But! There is a unique maximum!

- So, $F(\rho_A, \sigma_A) = \|\sqrt{\rho} \sqrt{\sigma}\|_1^2$ (by defⁿ).

$$\text{Also! } = \max_{\text{purifications}} |\langle \psi_\rho | \psi_\sigma \rangle|^2$$

This is Uhlmann's Theorem.

Review of Purification

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• recall: ① a pure state $|\psi\rangle$

② a mixed state ρ , s.t.: $\rho \geq 0$
 $\text{tr} \rho = 1$.

• can view a mixed state ρ_A in a H.S. A as arising from a pure state $|\psi\rangle_{RA}$ on an extended HS, $R \otimes A$,

$$\rho_{RA} = |\psi\rangle_{RA} \langle \psi|_{RA}.$$

• then discard system R and keep only A .

$|\psi\rangle_{RA}$ is the purification of ρ_A

→ this is not unique!

Different (infinite) $|\psi\rangle_{RA}$'s are possible that will give the same ρ_A after discarding R .

Construction:

• given a ρ_A , take eigendecompⁿ

$$\rho_A = \sum_i p_i |e_i\rangle \langle e_i|$$

• Define: $|\psi\rangle_{RA} = \sum_i \sqrt{p_i} |i\rangle_R |e_i\rangle_A$

→ $|i\rangle_R$ is an arbitrary ONB in another HS $\hat{c} \dim(R) = \dim(A)$.

• Check! Trace out system R by

applying: $\text{tr}_R \otimes I_A$
 $\uparrow \text{on } R \quad \uparrow \text{on } A$

on $|\psi\rangle_{RA} \langle \psi|_{RA}$

$$\begin{aligned} & \text{tr}_R \sum_{ij} \sqrt{p_i p_j} |i\rangle \langle j|_R \otimes |e_i\rangle \langle e_j|_A \\ & \quad \uparrow \quad \quad \quad \uparrow \\ & \quad \text{tr} \quad \quad \quad I_A \\ & = \sum_{ij} \sqrt{p_i p_j} \underbrace{\text{tr}(|i\rangle \langle j|_R)}_{= \delta_{ij}} \otimes |e_i\rangle \langle e_j|_A \\ & = \sum_i p_i |e_i\rangle \langle e_i|_A \\ & = \rho_A \quad \checkmark \end{aligned}$$

- Purifications are not unique

⑧

$$|\Psi_p\rangle_{RA} = \sum_i \sqrt{p_i} |i\rangle_R \otimes |e_i\rangle_A$$

← ONB on A as eigenvectors of ρ_A

← arbitrary ONB on another HS.

- we are free to rotate the ONB on R with a unitary U_R . That is,

$(U_R \otimes I_A) |\Psi_p\rangle_{RA}$ will also be a purification for ρ_A given a purification $|\Psi_p\rangle_{RA}$.

- relation to fidelity between states ρ & σ :

Given purifications: $\rho_A \longrightarrow |\Psi_\rho\rangle_{RA}$

$\sigma_A \longrightarrow |\Psi_\sigma\rangle_{RA}$

we can define a fidelity between ρ & σ as the overlap of the purifications:

$$|\langle \Psi_\rho | \Psi_\sigma \rangle|^2.$$

→ Define the fidelity btwn ρ & σ as:

$$F(\rho, \sigma) = \max_{\text{purifications}} |\langle \Psi_\rho | \Psi_\sigma \rangle|^2$$

→ Then show this also equals: $\|\sqrt{\rho} \sqrt{\sigma}\|_1^2$ as well

→ no optimization procedure required: A direct calcⁿ.

This is Uhlmann's Th^m.